Solving the Field: By Integration, Then by Gauss's Law

Three objects worked twice over — and the one the short way cannot touch

> WHERE THIS SITS
>
> **Note 6** established Gauss's law, $\varepsilon_0\Phi = q_{\text{enclosed}}$, and checked it on a point charge. This note spends it. Three charged objects are worked right through by **direct integration** first, and then the same objects are handed to Gauss's law. **Watch which ones come back cheap, and which one does not.**

The hard way first

Gauss's law is about to look like a cheat, so it is worth seeing the bill it is paying. Here are three charged objects done by **direct integration** — set up the element, write its distance, add up the contributions. Two of these three will come back in two lines each once Gauss is let loose on them. **The third will not, and that is the point of including it.**

A line of charge

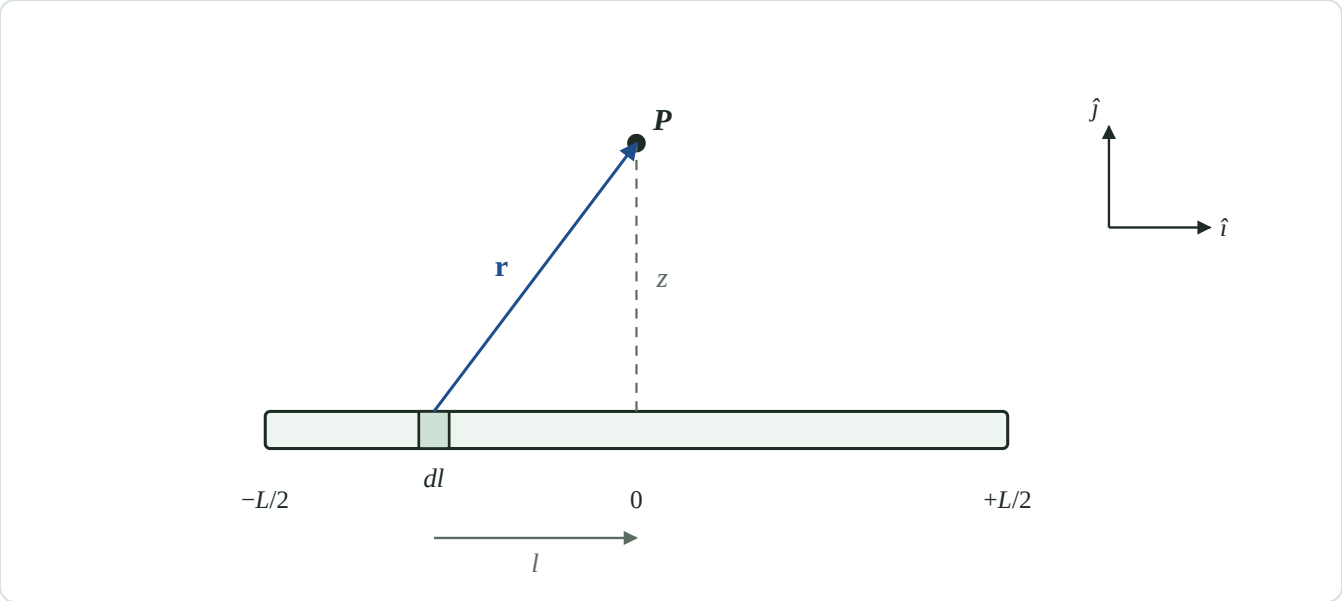


Figure 8. His construction: a rod from $-L/2$ to $+L/2$, the element dl at distance l from the centre, the field point P at height z. **Everything that follows is read off this picture**, which is why he draws it before writing anything.

Then he does what he does at the end of every one of these — he takes the limits and checks them.

A ring of charge

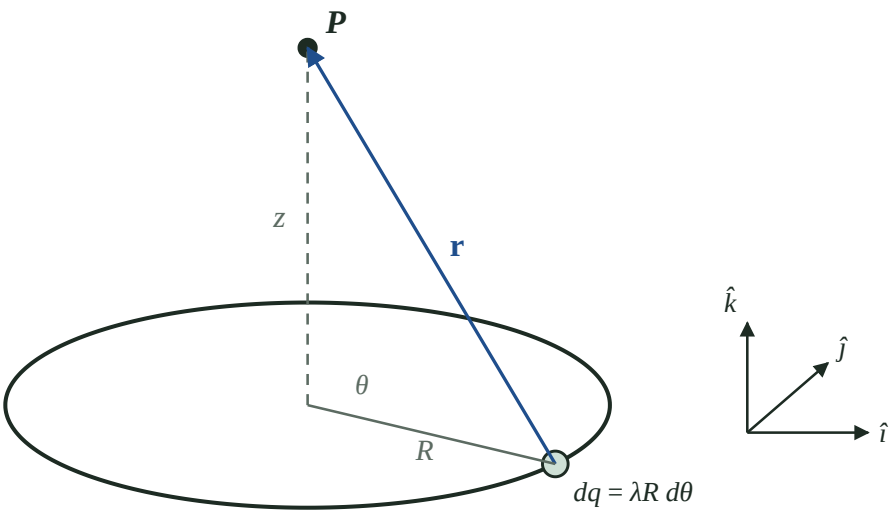


Figure 9. His construction for the ring, with the field point on the axis. Every element is **the same distance** from *P* — which is what makes this one doable at all.

A disc of charge

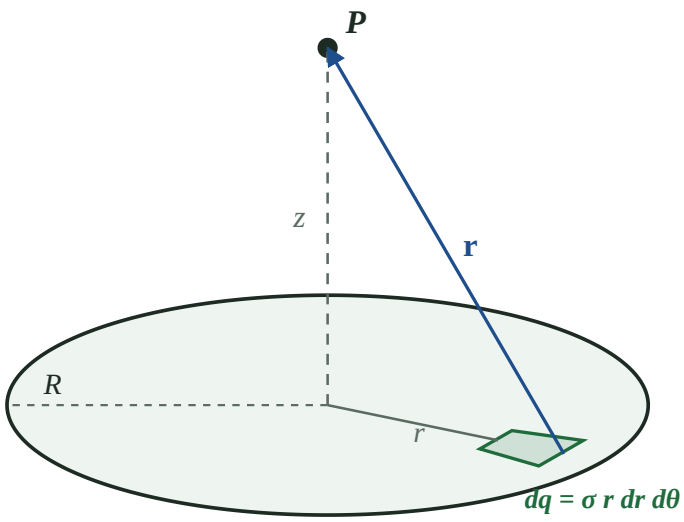


Figure 10. His construction for the disc. The element is a little wedge and the integral runs over both r and θ.

KEEP SCORE

Three objects, three pages of calculus, three answers: $\lambda/2\pi\varepsilon_0 z$ for the long line, $Qz/4\pi\varepsilon_0(R^2+z^2)^{3/2}$ for the ring, and $\sigma/2\varepsilon_0$ for the big flat sheet. **Watch which two come back.**

3. Solving the Electric Field with Gauss's Law: Examples

THE METHOD, IN FOUR STEPS

The previous note gave the two conditions; this is the procedure they make. Every example below runs these same four beats, and the ring at the end fails at step 2.

1. **Name the symmetry.** Spherical, cylindrical or planar — that is the whole list, and it decides everything after it.
2. **Draw the surface that symmetry hands you**: the one where **E** lies along the outward normal, and has **the same size everywhere it matters**. Those are his two conditions, and they are the only reason *E* can come out of the integral.
3. **Count q_{enc}** — what is inside that surface, and nothing else.
4. $\varepsilon_0\Phi = q_{enc}$, and solve for *E*.

A solid insulating sphere, charged all the way through

Spherical → a concentric sphere → $q_{enc} = \rho_0 \times$ the charged volume it encloses — which stops growing at $r = R$ → solve.

His annotations set this one up: **"object = insulator as opposed to conductor"** and **"charge is evenly distributed."** That matters, because it is what lets you say how much charge a surface drawn *inside* the object encloses.

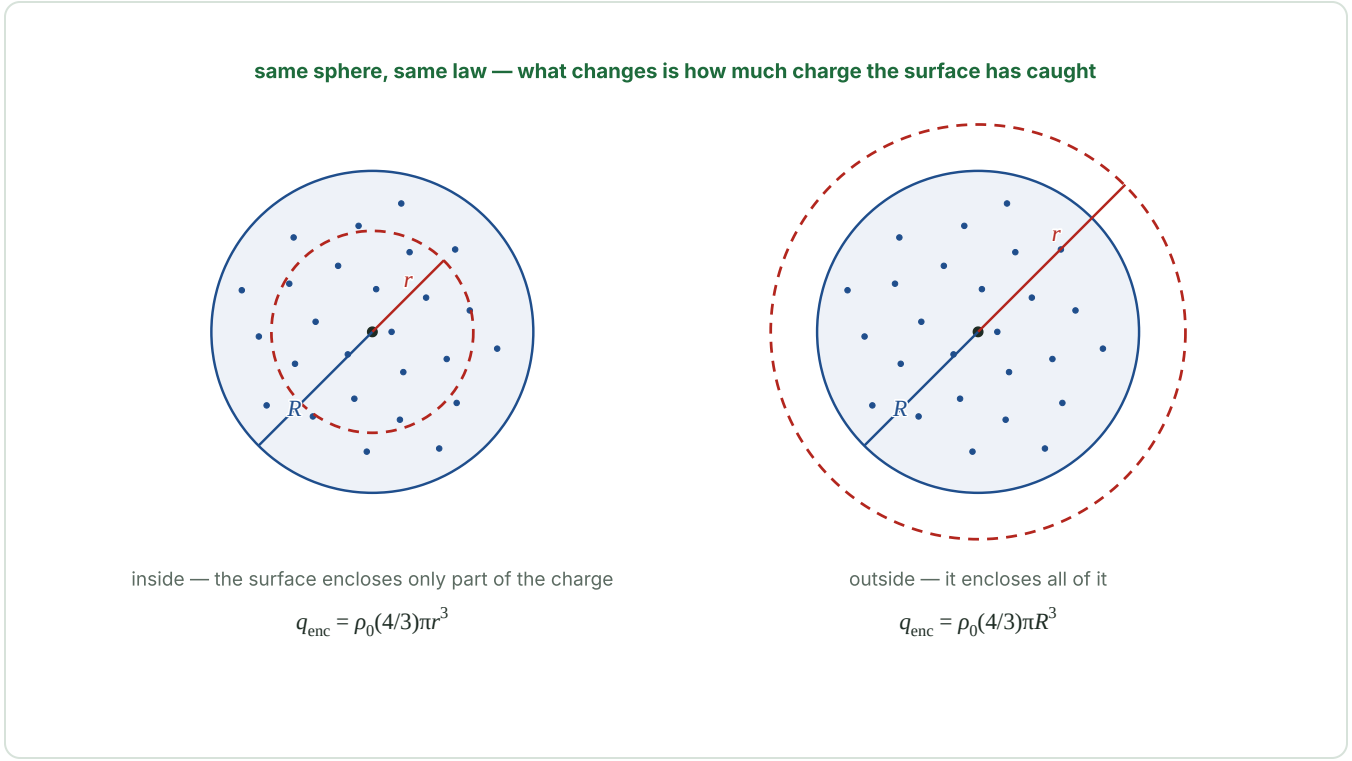


Figure 11. The two cases he draws side by side: a Gaussian surface inside the charged sphere, and one outside it. Inside, the surface encloses only part of the charge; outside, it encloses all of it.

Two different laws in the two regions, and his graph of them is the thing to carry away:

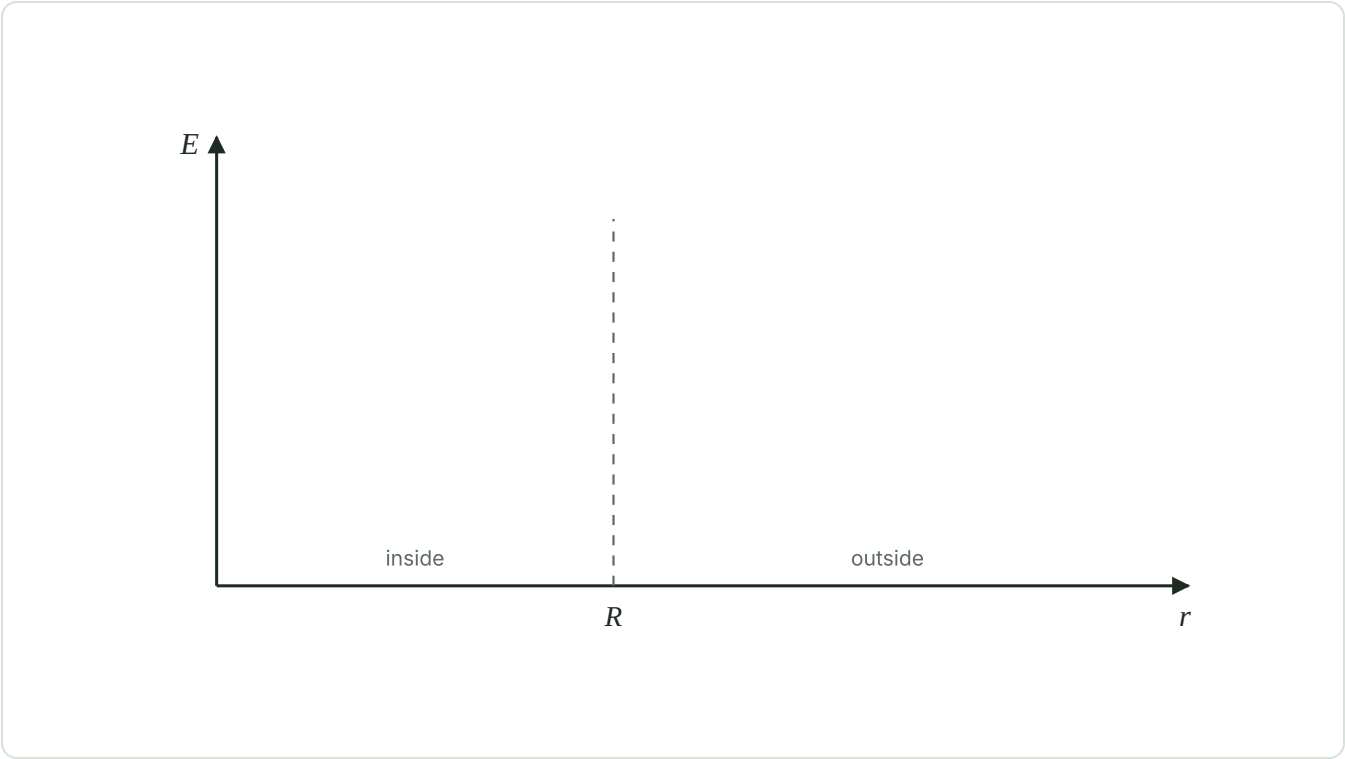
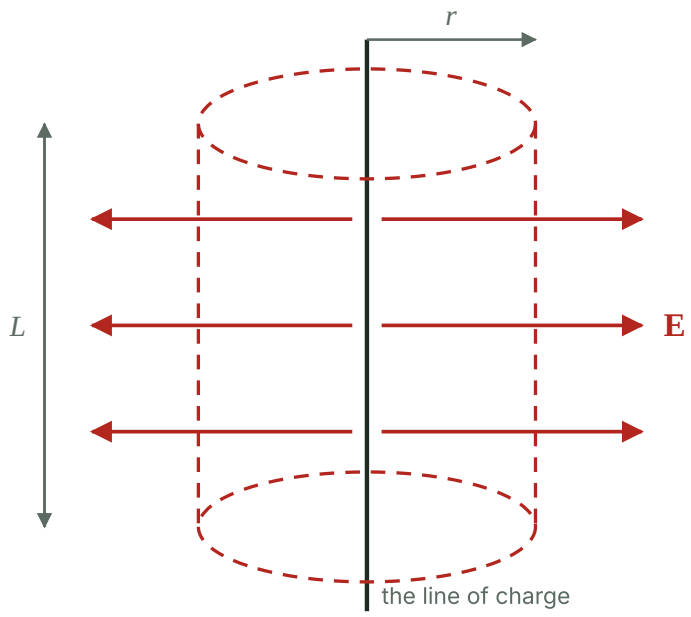


Figure 12. Plot both branches. Straight line out to R, then falling away as $1/r^2$ — and check that they meet at the surface.

A long line of charge — the one we already did the hard way

Cylindrical → a coaxial cylinder, no flux through the ends → $q_{\text{enc}} = \lambda L$ → solve.



no flux through the top or the bottom — the field is parallel to them

Figure 13. The Gaussian surface is a cylinder sharing the line's axis. His note on it: **"cylinder without top & bottom"** — not because they are missing, but because **no flux goes through them**.

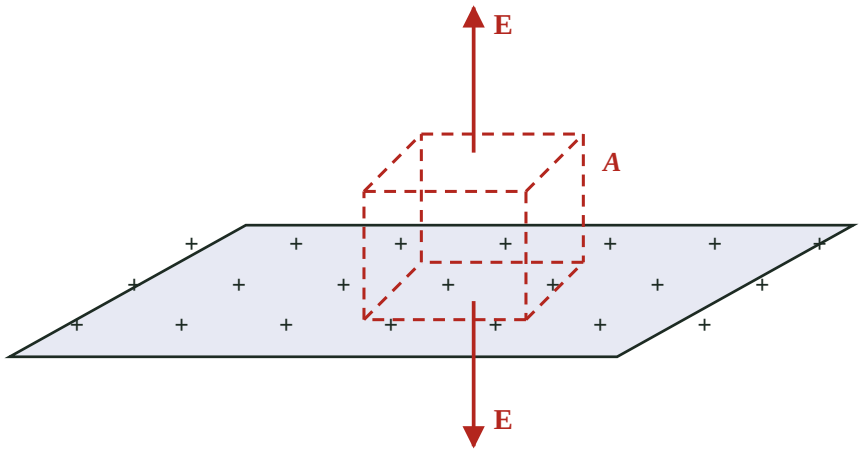
COMPARE

three lines against two pages

That is the same $\lambda/2\pi\varepsilon_0 r$ you got earlier from a vector integral, a Taylor expansion and a limit. **Nothing was skipped.** The symmetry that made the dot product trivial and pulled E out of the integral did all the work the calculus had been doing by hand.

A flat sheet of charge — also already done the hard way

Planar → a pillbox straddling the sheet, no flux through the sides → $q_{enc} = \sigma A$ → solve.



flux leaves through TWO faces, so the 2 in $E(2A)$ is the whole story

Figure 14. A box straddling the sheet. The sides carry no flux, so only the two end faces count — which is where the 2 comes from.

AND THE ONE THAT DID NOT COME BACK

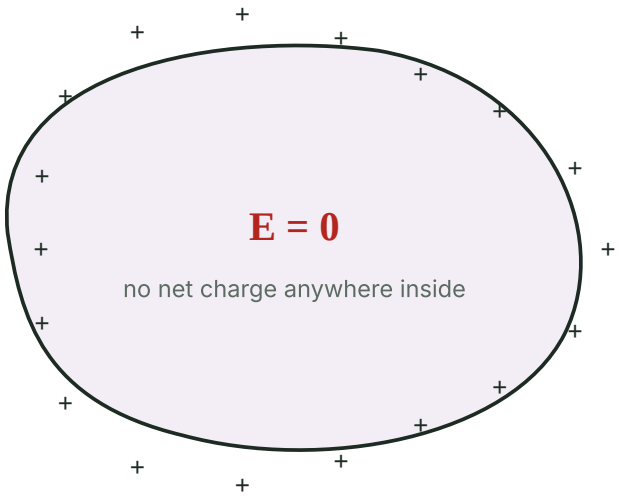
The ring fails at step 2. Try to draw a Gaussian surface for it and you will find there is no good one: no surface around a ring has the field at the same strength all over it, and none has the field everywhere along its normal. **Gauss's law is still true for the ring — it is always true — it is just useless**, because you cannot get *E* out of the integral. That is the honest boundary of the method, and it is why the ring was worth the page.

5. Charge on a Conductor

His opening annotation is the whole explanation: **"Conductor: free electrons!!"** and then, in full:

> "Electrons will respond immediately to electric field and redistribute to establish new equilibrium. e¯s no longer move!!"

So he asks the question that follows: **"Where do excess charge go?"**



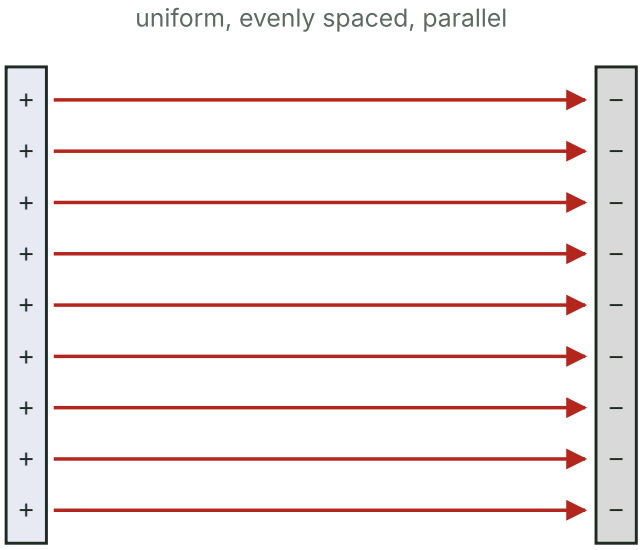
the electrons are free, so they move until nothing is pushing them — and that is the surface

Figure 15. His answer, written across the figure: **"No net charge interior, *E* = 0"** and **"They go to the surface!!"** **Note the shape** — he draws a lump, not a sphere, because the result does not need one.

One consequence he flags with **"Note!!"**, and then uses:

$E = \sigma / 2\varepsilon_0$ — one sheet, from the pillbox above

$E = \sigma / \varepsilon_0$ — between two oppositely charged plates



$$E = \sigma / \varepsilon_0$$

Figure 16. Two plates, and his result for the gap, $E = \sigma/\varepsilon_0$ — twice one sheet's, because **both** sheets push a positive test charge the same way in the gap, and opposite ways outside it.

ONE TO CHECK FOR YOURSELF

On his page the first of those two figures is drawn as a **conductor**, with $E = 0$ on one side, and the annotation beside it reads $\sigma/2\varepsilon_0$. Work out what the pillbox gives you for a **conducting surface**, where all the flux leaves through one face instead of two, and see whether you get the same thing. **Then ask which of the two pictures the annotation was meant for.**

This is also where **Note 5** picks the thread up: every geometry on this page has a potential to go with it, got by integrating the field you just found. $V(r) = -\int \mathbf{E}\cdot d\mathbf{r}$ turns the sphere, the shell and the cylinder here into the potentials worked there — **same objects, same hand, other quantity**.

THE WHOLE NOTE IN ONE LINE

$$\varepsilon_0 \Phi = q_{\text{enclosed}}$$

Gauss's law is **always true** and **only sometimes useful**. It is true for the ring, for a lump of putty, for your hand — and it is useful exactly when the object has enough symmetry that you can find a surface where E is constant and along the normal. Sphere, cylinder, plane. That is the list.

Looking ahead

That is the field half of Module 3 finished. **Note 12** spends the potential side of it on real problems — alpha decay, how hard fusion is, and a pair of plates used as an accelerator — all of which are bookkeeping between U and K once you have the potentials this module has built.